

SUNDAY IKECHUKWU

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RESEARCH INTEREST

Robot Learning & Adaptive Control • Imitation and Reinforcement Learning • Safe Autonomy • Continual Adaptation in Unstructured Environments.

EDUCATION

University of Benin, Benin City, Edo State, Nigeria

Jan. 2020 - Jan. 2025

Bachelor of Engineering (B.Eng) in Mechatronics Engineering

Honors: First Class Honors (Distinction), Class Rank: 1 out of 120

CGPA: 4.61/5.0 (WES iGPA: 3.77/4.0 | Scholaro: 3.796/4.0)

Thesis Title: Design and Implementation of a Microcontroller-Based Biometric Authentication System for E-Voting

RESEARCH EXPERIENCE

Graduate Research & Teaching Assistant

June 2026 – Present

Machine Intelligence Research Group (MIRG) University of Lagos | *Research Lead: Dr. B.A Sawyerr*

- Collaborating with the Research Lead on advanced initiatives within Artificial Intelligence, Machine Learning, and Optimization algorithms.
- Preparing an independent research framework on deterministic safety shields for autonomous navigation for co-authored submission to ICAIR 2026.
- Facilitating undergraduate Computer Science education by evaluating assignments and grading examinations.

Graduate Research Assistant, University of Benin

May 2025 – Present

Supervisor: Prof. Ariavie G.O.

- **Project 1: "Towards Automated Maritime Logistics: Simulation-Based Design of a Robotic Arm for Container Ship Cargo Handling" (Mechatronics Lab)**
 - Engineered a distributed, multi-script control architecture in Lua, synchronizing a UR10 6-DOF robotic arm with conveyor actuators and spawn triggers for a fully autonomous pick-place-return cycle.
 - Developed staged, sensor-driven control logic to dynamically regulate conveyor flow; implemented anti-stacking and collision-avoidance protocols to optimize throughput and prevent pick-pose overcrowding.
 - Integrated CoppeliaSim's simIK module for robust trajectory generation and motion planning, paired with an algorithmic 5x4 grid spatial mapping for precise, automated cargo palletizing.
- **Project 2: "A Vision-Based Approach to Mitigating Vandalism and Enhancing Leak Detection in Nigeria's Oil and Gas Industry"**
 - Developing a real-time anomaly detection pipeline using YOLO and thermal imaging for deployment on edge devices (UAVs)
 - Conducted a gap analysis of 50+ publications to optimize the architecture for low-latency inference in field operations.
 - Leading custom dataset creation and validation protocols, supporting model training, real-world testing, and preparation of research findings for publication.

Graduate Research Assistant – Computer Vision for Critical Infrastructure Monitoring

Jul. 2025 – Sept. 2025

Principal Investigator: Dr. Isi Edeoghon

University of Benin

Research Focus: Deep Learning frameworks for autonomous fault detection in electrical infrastructure, addressing critical challenges in preventive maintenance and grid reliability

- Designed and implemented a two-stage deep learning pipeline combining Mask R-CNN (instance segmentation) with hierarchical classification for precise fault localization in complex aerial imagery.
- Solved the challenge of class confusion in cluttered backgrounds by implementing targeted data augmentation protocols, directly improving mean Average Precision (mAP) scores
- Engineered the system to be robust against real-world complexities, specifically occlusions and illumination variations in aerial imagery.

Undergraduate Final Year Thesis

Jul. 2024 – Feb. 2025

University of Benin Mechatronics Lab | Supervisor: Prof. Obanor A.I.

Thesis Title: Design and Implementation of a Microcontroller-Based Biometric Authentication System for E-Voting

- Led a 4-member team to research vulnerabilities in Nigeria's electoral process to conceptualize a technology-driven solution for enhancing transparency and trust.
- Designed and developed a proof-of-concept electronic voting system with fingerprint-based biometric authentication for secure voter identification.
- Validated the system through simulation, integrating custom hardware (Arduino/microcontrollers) with software for real-time data processing.

PUBLICATIONS & MANUSCRIPTS

- **Ikechukwu S.**, Sawyerr, B.A. "Shielded DAgger: A Deterministic Fallback Architecture for Safe Imitation Learning in Robotics" (Under Review).
- Ariavie, G.O., Simon, T. A., & **Ikechukwu S.** A Vision-Based Approach to Mitigating Vandalism and Enhancing Leak Detection in Nigeria's Oil and Gas Industry. (Manuscript in preparation).
- Ojefia, F. E., Arigbanla, A. A., **Ikechukwu S.**, Adetunji, E. O., Enodiana, O. M., & Ekwonu, F. T. (2025). Design of a Mini-Autonomous Vehicle for Smart Mobility Solutions. *World Journal of Advanced Research and Reviews*, 28(1), 434–448. <https://doi.org/10.30574/wjarr.2025.28.1.3443>

SELECTED ROBOTICS PROJECTS

Shielded Autonomy Architecture for Safe AI Navigation | *Independent Research* | [[GitHub](#)] | [[Project Page](#)] | **Apr. 2026 – Present**

Objective: Designed an end-to-end imitation learning pipeline featuring a deterministic safety shield to preemptively block catastrophic out-of-distribution collisions.

- Architected a real-time (10Hz) Safety Shield in ROS 2 that mathematically evaluates kinematic divergence ($\Delta v, \Delta w$) and costmap breaches between an end-to-end Neural Network and a deterministic MPPI expert controller, seamlessly triggering expert takeovers to prevent critical policy failures.
- Resolved Covariate Shift in Behavioral Cloning by deploying a custom Safe DAgger (Dataset Aggregation) pipeline, autonomously recording and oversampling critical edge-case recovery maneuvers to cure policy mean-regression.
- Engineered robust state-machine logic and dynamic pathing, utilizing continuous tf2 spatial transformations for 1.5m sub-goal tracking and implementing an "inactivity watchdog" to eliminate arbitration chattering during complex kinematic maneuvers
- Developed automated HDF5 data pipelines in Python/NumPy to programmatically purge physical deadlocks and rotational noise, ensuring high-purity datasets for PyTorch model training and low-latency ONNX inference.

UR5 Imitation Learning for Robotic Pick-and-Place | *Independent Project* | [[GitHub](#)] | [[Project Page](#)] | **May 2026 – June 2026**

- Implemented and evaluated Behavioral Cloning and Action Chunking Transformer (ACT) policies for UR5 robotic pick-and-place manipulation in CoppeliaSim, achieving 100% task success with ACT vs 0% with BC across 20 controlled evaluation episodes.
- Designed an automated expert demonstration pipeline using CoppeliaSim's ZMQ Remote API, collecting 100 HDF5-formatted episodes with full observation-action schema documentation and dataset analysis.
- Diagnosed and resolved a ZMQ thread-safety constraint, effective sampling rate mismatch, and joint control mode inconsistency during closed-loop inference—demonstrating systematic robotics systems debugging.
- Produced empirical evidence of the teacher forcing gap: BC achieved validation MSE 0.001 but 0% closed-loop success, motivating transition to transformer-based action chunking with documented failure mode analysis

TEACHING/ACADEMIC EXPERIENCE

Teaching Assistant, Department of Mechatronics Engineering

Jul 2025 – Present

University of Benin, Benin City, Edo State, Nigeria

- **Technical Evaluation:** Assessing and grading technical logbooks and industrial reports for final-year (500L) students returning from the Student Industrial Work Experience Scheme (SIWES), ensuring alignment with engineering standards
- **Examination Assessment:** Assisted Prof. G.O. Ariavie in grading examinations for core mechatronics course, ensuring accurate and consistent assessment of student performance for 100+ undergraduate students.
- Facilitated laboratory sessions for undergraduate students, providing guidance on robotics simulation tools and control system fundamentals.

Technical Instructor, Chestnut and Bolt Technology, Benin City Nigeria

Nov. 2024 – Jan. 2025

- Equipped interns with foundational skills in electronics, Embedded C, and Arduino IDE through hands-on training. Enabling them to transition from theoretical knowledge to practical prototyping projects within weeks.

Undergraduate Tutor, University of Benin, Benin City, Edo State, Nigeria

Jan. 2021 – Dec. 2024

- Provided peer-tutoring to over 70 undergraduate students in core courses including “Robotics and Automation”, Introduction to robotics, Control Engineering, helping ~80% of participating students secure “A” grades in these challenging courses.

PROFESSIONAL EXPERIENCE

Embedded System Intern, Chestnut & Bolt Technologies | Edo State, Nigeria

Jan. 2024 - Jun. 2024

- Designed and deployed a prototype pick-and-place robotic arm system, integrating PIC/Arduino microcontrollers with custom hardware.
- **Critical Contribution:** Diagnosed a "Sim-to-Real" reliability gap where minor variations in lighting and object friction caused grasping failures in physical deployment despite perfect simulation performance.
- Designed and simulated embedded control circuits in Proteus to validate system logic prior to physical prototyping.
- Collaborated with the engineering team to automate control systems, enhancing practical skills in system integration and component selection.

SKILLS

Technical Skills

- **Robotics & Simulation:** ROS2 (Nav2, SLAM), MoveIt, Coppeliasim, Gazebo, RViz Linux(Ubuntu)
- **AI & Computer Vision:** PyTorch, Roboflow, Mask R-CNN, Reinforcement Learning, Google Colab
- **Programming:** Python, Embedded C (Intermediate), Arduino Programing
- **Hardware & Electronics:** Arduino, PIC Controller(Basic), Circuit Design & Simulation using Proteus, soldering components onto PCBs

Soft Skills: Problem-Solving and Critical Thinking, Analytical Ability, Leadership and Teamwork, Communication and Interpersonal Skills, Time Management, Adaptability/Flexibility, Inquisitiveness, Continuous Learning.

PRESENTATION

- Design and Implementation of a Microcontroller-Based Biometric Authentication System for E-Voting. Undergraduate Thesis Defense, Department of Mechatronics Engineering, University of Benin, 2023.
- Plastic Waste Pollution in Nigeria. Technical presentation for MTE 381: Technical Report Writing & Presentation, University of Benin, 2021

LEADERSHIP & COMMUNITY ENGAGEMENT

COMMUNITY DEVELOPMENT VOLUNTEER

Jun. 2026 – Present

Social Welfare & Tourism CDS Group | Lagos, Nigeria

- Spearheading public health and wellness initiatives targeting underserved demographic groups, including organizing the "Health, Rest & Everyday Wellness Outreach" at Makoko Market.
- Educating locals on preventative healthcare, occupational stress management, and practical lifestyle habits to improve long-term well-being.
- Collaborating with a team of fellow graduates to design and execute grassroots community interventions that promote healthier living standards.

Science & Math Club Mentor (Volunteer)- Chrisidia Education Center, Benin City.

05/2025 – 07/2025

- Guided the school’s Science and Math Club, introducing over 300 students to innovative STEM concepts beyond the classroom.
- Organized mathematics quizzes and collaborative problem-solving sessions that boosted student engagement and performance.
- Introduced robotics and Arduino programming through hands-on demonstrations, sparking early interest in engineering and technology among high school students.

Academic Mentorship, University of Benin

05/2023 – Present

- Directed two undergraduate researchers in virtual prototyping, teaching Coppeliasim simulation best practices, distributed control logic, and motion planning strategies.
- Mentored junior undergraduate students in the Mechatronics Engineering program, providing guidance on effective approaches to challenging core courses and study strategies.
- Made course materials and academic support consistently available, resulting in measurable performance improvements, with several mentees now on track to graduate with First Class honors.

SECRETARY GENERAL, Mechatronics Students Association, University of Benin

03/2024 – 10/2024

- Coordinated operations, managed records, and organized academic workshops, strengthening collaboration among engineering students.

MEMBER OF PARLIAMENT, National Association of Marine, Mechanical and Mechatronics Engineering Students (NAMES)- UNIBEN Chapter, University of Benin

03/2024 – 10/2024

- Represented mechatronics students in parliamentary sessions, contributing to policy discussions and decision-making on student welfare and academic matters.

AWARDS AND HONOURS

• Best Graduating Student, Department of Mechatronics Engineering, University of Benin	Apr. 2025
• First-Ever First Class Graduate, Department of Mechatronics Engineering University of Benin	Apr. 2025
• Best Undergraduate Final Year Research Thesis, Mechatronics Engineering	Feb. 2025
• Best 300Level Student, Department of Mechatronics Engineering, University of Benin	Nov. 2023
• Monday Onyeme Foundation Scholarship Award	Sep. 2023
• Best Technical Report in MTE381(Technical Report Writing & Presentation).	May 2023
• TotalEnergies/JDA Merit Scholarship	May 2022
• Asiruwa Scholarship Foundation for Outstanding Academic Performance	Jan. 2022
• Best 100Level Student across M3(Marine, Mechanical and Mechatronics) Engineering Departments	Sep. 2021

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS

• UC Berkeley CS 285: Deep Reinforcement Learning	<i>In Progress</i>
• Deep Reinforcement Learning <i>Hugging Face</i>	<i>In Progress</i>
• Modern Robotics, Course 2: Robot Kinematics <i>Coursera (Northwestern University)</i>	<i>Oct. 2025</i>
• Modern Robotics, Course 1: Foundations of Robot Motion <i>Coursera (Northwestern University)</i>	<i>Sep. 2025</i>
• Fundamentals of Manuscript Preparation <i>Elsevier Research Academy</i>	<i>Sep. 2025</i>
• Data Analysis with Python <i>freeCodeCamp</i>	<i>Mar. 2025</i>
• Supervised Machine Learning: Regression and Classification <i>Coursera (Stanford University)</i>	<i>Apr. 2024</i>

PROFESSIONAL MEMBERSHIPS

- Black in Robotics BiR
- Regular Member, African Robotics Network AfROB
- Nigeria Institute of Professional Engineers and Scientists, NIPES.
- Nigeria Institute of Mechanical Engineers, NiMechE.